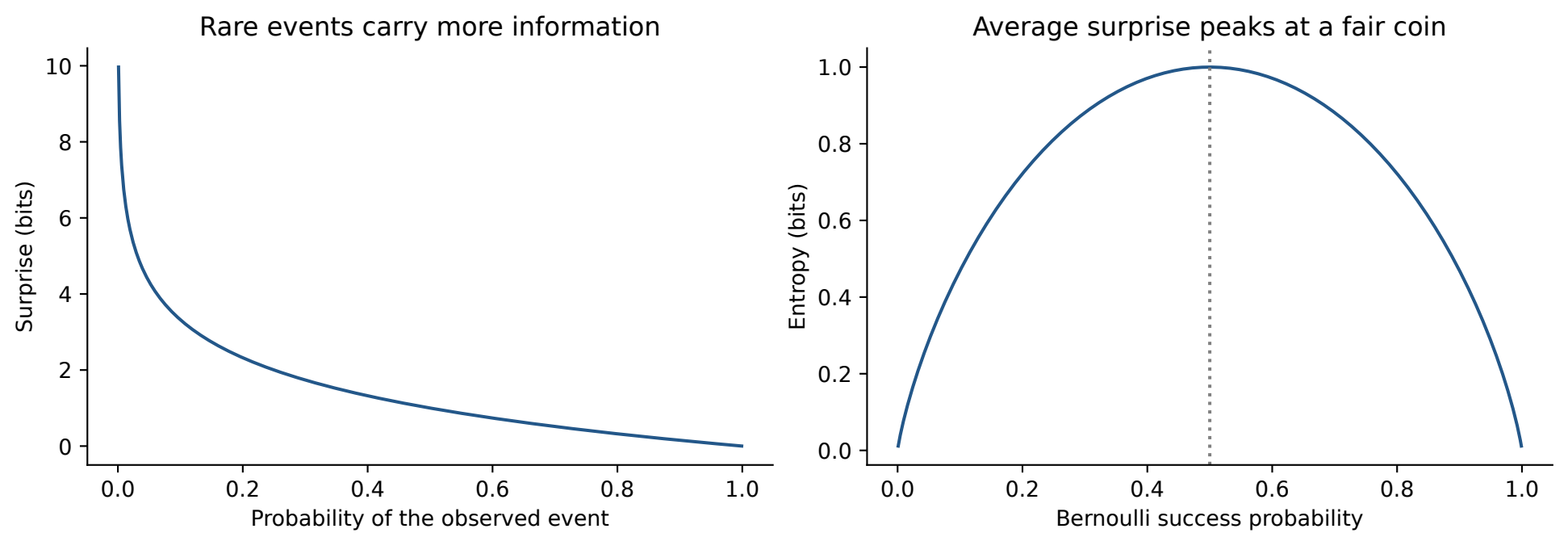
Rare events carry more information
Surprise (bits)
10
8
6
4
2
0
0.0
0.2
0.4
0.6
0.8
1.0
Probability of the observed event
Average surprise peaks at a fair coin
Entropy (bits)
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
Bernoulli success probability